Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Properties of Cubes and Cuboids

A three-dimensional shape is solid, so it has three measures: length, breadth (width) and height. When we study a solid shape, we look at its faces, its edges and its vertices (corners). Learning these three things for a shape lets you recognise it and tell it apart from other shapes.

A face is a flat surface of the shape. An edge is a line where two faces meet. A vertex (plural: vertices) is a corner, where edges meet.

Let's look at a cuboid first. A cuboid is a box whose faces are rectangles — think of a brick, a book or a matchbox. Every cuboid has the same properties:

- Faces: 6
- Edges: 12
- Vertices: 8

Notice that not all the faces of a cuboid look the same, because a cuboid's faces are rectangles of different sizes. But it still always has exactly 6 faces, 12 edges and 8 vertices.

Now look at a cube. A cube is a special cuboid: every one of its faces is a square, and all the squares are the same size. Think of a dice. Because a cube is just a cuboid with equal square faces, it has the same counts:

- Faces: 6
- Edges: 12
- Vertices: 8

So both a cube and a cuboid have 6 faces, 12 edges and 8 vertices. The difference between them is the shape of their faces: a cube's 6 faces are all equal squares, while a cuboid's faces are rectangles (not all the same).

Let's work through the properties of a cube, then a cuboid.

A cube has 6 faces. Count them: front, back, top, bottom, left and right — that is 6. Each of these faces is a square, and they are all the same size.

A cube has 12 edges. The front face has 4 edges, and the back face has 4 edges, and there are 4 edges joining the front to the back — $4 + 4 + 4 = 12$.

A cube has 8 vertices — one at each corner, top and bottom, front and back: 4 on top and 4 on the bottom.

A cuboid has exactly the same counts: 6 faces, 12 edges, 8 vertices. The only difference is the shape of its faces. A cuboid's faces are rectangles (for example a front that is taller than it is wide), and opposite faces are the same size. In a cube all six faces are equal squares.

So when you are asked for the properties of a cube or a cuboid, answer with the three counts — faces, edges and vertices — and, for a cube, add that all its faces are equal squares.

Practice

1. How many vertices has a cuboid?
2. How many edges has a cube?
3. A box of chalk is the shape of a cuboid. List its number of faces, edges and vertices.
4. A dice is in the shape of a cube. How many faces does it have, and what shape is each face?

Section 2: Pyramids and Cones

In the last section you looked at cubes and cuboids — shapes built from flat square and rectangular faces. Now meet two more three-dimensional figures: pyramids and cones. Both are built from a flat base that rises up to a single point, but their sides are very different.

A pyramid has a flat base that is a polygon (a triangle, a square, a pentagon, a hexagon, and so on), and every other face is a triangle. All these triangular faces meet at one point at the top, called the apex (or vertex). A pyramid is named by the shape of its base: a pyramid with a square base is a square pyramid, one with a triangular base is a triangular pyramid, and one with a hexagonal base is a hexagonal pyramid.

A cone has a flat circular base, and instead of triangular faces it has one curved surface that tapers smoothly up to a single point, the apex. So while a pyramid is made entirely of flat faces, a cone has exactly one curved face.

Let's list their properties side by side.

Here are the properties to know for each figure.

Square pyramid (a pyramid with a square base):

It has 5 faces: 1 square base and 4 triangular faces.
It has 8 edges.
It has 5 vertices: the 4 corners of the square base plus the apex.

Hexagonal pyramid (a pyramid with a hexagonal base):

It has 7 faces: 1 hexagonal base and 6 triangular faces.
It has 12 edges.
It has 7 vertices: the 6 corners of the hexagon base plus the apex.

Cone:

It has 2 faces: 1 circular base and 1 curved surface.
It has 0 edges (no straight edges at all).
It has 1 vertex: the apex.

Spot the pattern for a pyramid: if the base is a polygon with n sides, the pyramid has n triangular faces plus the 1 base (so n + 1 faces in total), n + 1 vertices, and 2n edges. Check it against the square pyramid: n = 4, so faces = 5, vertices = 5, edges = 8. That works! And the hexagon: n = 6, so faces = 7, vertices = 7, edges = 12.

Worked example — listing the properties of a pentagonal pyramid

A pentagonal pyramid has a pentagon (5 sides) for its base.

Faces. It has 1 pentagonal base and 5 triangular faces that rise to the apex. So the number of faces is 5 + 1 = 6.

Vertices. It has the 5 corners of the base plus the 1 apex, so 5 + 1 = 6 vertices.

Edges. Using the pattern, edges = 2 × n = 2 × 5 = 10.

Check by counting: the base is a pentagon (5 edges), and each of the 5 corners sends an edge up to the apex (5 more edges). That gives 5 + 5 = 10 edges.

So a pentagonal pyramid has 6 faces, 6 vertices and 10 edges.

Practice

1. A hexagonal pyramid has a hexagon for its base. List all its properties: how many faces, how many vertices and how many edges does it have?
2. A tetrahedron is a pyramid with a triangular base. How many vertices does a tetrahedron have?
3. A cone has how many faces, edges and vertices?

Check your work

Attempt every question before you read this page.

Properties of Cubes and Cuboids

1. A cuboid has 8 vertices (corners).

   A cuboid has 4 corners at the top and 4 corners at the bottom, so $4 + 4 = 8$ vertices in total.
2. A cube has 12 edges.

   A cube has 4 edges on the top face, 4 edges on the bottom face, and 4 vertical edges joining the top to the bottom. $4 + 4 + 4 = 12$ edges.
3. A cuboid (the chalk box) has 6 faces, 12 edges and 8 vertices.

   Every cuboid — a chalk box, a book or a brick — has the same three counts: 6 faces, 12 edges and 8 vertices.
4. A cube has 6 faces, and every face is a square (all the same size).

   A cube is a cuboid whose 6 faces are all equal squares, which is why a dice — a cube — has 6 square faces.

Pyramids and Cones

1. 7 faces (1 hexagonal base and 6 triangular faces), 7 vertices (6 base corners plus the apex), and 12 edges.

   The base is a hexagon, so $n = 6$. Faces $= n + 1 = 7$. Vertices $= n + 1 = 7$. Edges $= 2n = 2 \times 6 = 12$.
2. 4 vertices: the 3 corners of the triangular base plus the 1 apex.

   $n = 3$ for a triangle. Vertices $= n + 1 = 3 + 1 = 4$.
3. 2 faces (1 circular base and 1 curved surface), 0 edges, and 1 vertex (the apex).

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-df6d256a	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____